

Integrating Landsat, Sentinel-2 and Sentinel-1 time series for mapping intermediate crops

Kassandra Jenssch^a, Gohar Ghazaryan^{b,a}, Stefan Ernst^a, Patrick Hostert^{a,c} and Claas Nendel^{b,d,e,c}

^aGeography Department, Humboldt-Universität zu Berlin, Berlin, Germany; ^bLeibniz Centre for Agricultural Landscape Research (ZALF), Münchenberg, Germany; ^cIntegrative Research Institute on Transformations of Human-Environment Systems (IRI THESys), Humboldt-Universität zu Berlin, Berlin, Germany; ^dInstitute of Biochemistry and Biology, University of Potsdam, Potsdam, Germany; ^eGlobal Change Research Institute CAS, Brno, Czech Republic

ABSTRACT

Intermediate crops are grown between main crops to protect soils and nutrients when fields would otherwise be bare. Despite being an essential constituent of cropping systems, spatial information on intermediate crops is scarce. Here, we propose a classification algorithm that combines field data, satellite imagery from multiple optical sensors and synthetic-aperture radar (SAR) data to map intermediate crops across Brandenburg, Germany. We trained random forest models using different sets of input features, including spectral-temporal metrics from optical data, metrics derived from SAR data and information on the scheduled main crop. The best classification was based on a combination of all input features and achieved an overall accuracy of 92.9%. Intermediate crops were overestimated, which can be partly attributed to misclassification of volunteers and weeds as intermediate crops. The overestimation was mitigated by aggregating results to the field level. Our results highlight the need for good optical data coverage during autumn and winter to accurately map intermediate crops while demonstrating the ability of SAR data to enhance classification accuracy. Overall, our study shows the potential of remote sensing methods to capture the characteristics of intermediate crops and derive spatially explicit data for monitoring sustainable agricultural practices.

ARTICLE HISTORY

Received 6 October 2024

Revised 28 March 2025

Accepted 13 May 2025

KEYWORDS

Cover crops; catch crops; agriculture monitoring; land cover classification; random forest; spectral-temporal metrics

Introduction

Current agricultural landscapes are dominated by simplified cropping systems that evolved from agricultural intensification. These systems are characterised by fewer species in rotations, enabling farmers to focus on the economically most important crops and enhancing agricultural productivity (Hufnagel et al., 2020). However, they rely on intensive management practices and external inputs that may cause nutrient leaching (Bowles et al., 2018), soil erosion (Montgomery, 2007), climate-relevant greenhouse gas emissions (Bowles et al., 2018; Smith et al., 2008) and a decline in biodiversity (Tscharrntke et al., 2012). On the contrary, more diverse cropping systems are likely to maintain productivity and resilience while reducing adverse impacts on the environment (Renard & Tilman, 2019; Rosa-Schleich et al., 2019). Crop diversification therefore holds significant potential for addressing environmental challenges and has become an increasingly discussed topic in science and policy to enhance agricultural sustainability.

The integration of Intermediate Crops (ICs) into rotations is one important diversification strategy that especially targets soil health. ICs are annual plants that are grown during fallow periods in summer or during

winter, but not for the purpose of producing any marketable products (Eurostat, 2023). Being commonly known as cover, catch or break crops, the term 'intermediate crops' addresses them as an umbrella term. It reflects their function in the system and the terms used in most languages in Europe, other than English. Generally, ICs may be used as fodder or for biogas production, but their primary function depends on the condition of the respective field. While the numerous environmental benefits of ICs are recognised through targeted agricultural policies, spatial information on IC cultivation is scarce. In this study, we evaluate the potential of integrating various remote sensing data for mapping ICs in north-eastern Germany. We aim to address challenges such as data gaps during winter by combining data from different sensors and to map ICs as comprehensively as possible by using field data. The field data were collected in a campaign tailored to the classification problem, particularly addressing crop types that could be confused with ICs due to similar growing patterns, and that are not included in official campaigns. This allows us to cover a broad range of ICs (regarding different growing periods, species compositions and functions)

CONTACT Gohar Ghazaryan ✉ gohar.ghazaryan@zalf.de Leibniz Centre for Agricultural Landscape Research (ZALF), Eberswalder Str. 84, 15374 Münchenberg, Germany

 Supplemental data for this article can be accessed online at <https://doi.org/10.1080/22797254.2025.2507738>

© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.

This is an Open Access article distributed under the terms of the Creative Commons Attribution License (<http://creativecommons.org/licenses/by/4.0/>), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

reflecting real-world agricultural practices across the study area. As a consequence, the study does not focus on mapping specific IC types but rather aims to represent the diversity of crops cultivated as ICs.

A great variety of species can be cultivated as ICs, either in mixtures or as sole crops. ICs can also be grown simultaneously with agronomic crops, i.e. as an undersown crop that remains on the field after the main crop was harvested. Ensuring a more continuous plant cover on agricultural soils, ICs help to control soil erosion (“cover crops”; Langdale et al. (1991)) and prevent nutrient leaching (Thapa et al., 2018). Certain ICs are specifically grown to enhance soil fertility (“fertility-building crops”), either through fixing atmospheric nitrogen in root-colonising bacteria and providing it to the succeeding crop (Dabney et al., 2011) or through leaving above- and belowground residues that increase soil organic matter in the long term (Poeplau & Don, 2015). Some ICs even improve the physical conditions in the soil through strong, deep-growing roots that may alleviate soil compaction (“conditioning crops”; Chen and Weil (2010)). Deep rooting ICs tap nitrogen reservoirs that have accumulated below the root zone of the main crops and bring this nitrogen back to the soil surface for later re-use by the main crops in the rotation (“catch crops”; Thorup-Kristensen et al. (2020)). ICs may have phytosanitary effects and break the cycle of soil-borne pests or diseases (“break crops”; Kirkegaard et al. (2008)) or suppress weeds through vigorous growth (“smother crops”; Liebman and Davis (2000)). Furthermore, ICs provide late flowering and contribute to more diverse habitats supporting soil organisms, birds and insects (Blanco-Canqui et al., 2015).

In the European Union (EU), the cultivation of ICs has been financially supported since the 2013 reform of the Common Agricultural Policy (CAP). Until the beginning of 2023, ICs were partly eligible for subsidies related to two of the three greening measures – crop diversification and ecological focus areas (Regulation 1307/2013). With the 2021 CAP reform, ICs have become an integral component of the ruleset contributing to good agricultural and environmental conditions (GAECs), and therefore to conditionality (Regulation 2021/2115). ICs cultivated with the intention of receiving direct payments would thus be reported by farmers and included in the Integrated Administration and Control System (IACS) of the EU (Regulation 2021/2116). The IACS provides information for almost every field in Europe, detailing the main crop grown during the reporting period (May) of each year and whether an IC eligible for subsidies is cultivated. This self-reporting by farmers is, however, also the weakness of the data, as it is neither complete, nor completely reliable. It requires independent on-site controls to validate the farmers’ reports. Notably, the system does not fully capture the complexity of cultivated ICs, as some farmers do not apply for

subsidy payments and not all ICs qualified for subsidies under greening measures in certain member states, e.g. Germany (DirektZahlDurchfV, 2014, § 31).

Spatial information on the cultivation of ICs thus remains scarce and incomplete, although it is of critical importance for assessing the efficacy of policy measures and environmental implications. For such assessments, mechanistic (i.e. process-based) agro-ecosystem models are often used, as they provide a relatively simple, quick and low-cost method to investigate the impact of climate or management changes on productivity and environment (Nendel et al., 2014). Such models simulate yields, nitrate leaching, carbon storage and greenhouse gas emissions in response to weather, site conditions and management. Applying them at larger scales requires spatial data to inform the models, and simulations that specifically require information on ICs could benefit from a more detailed monitoring of crop rotation patterns (Faye et al., 2023; Kollas et al., 2015; Nendel et al., 2023).

Providing information from satellite remote sensing is the main method to approach this objective, as it can efficiently capture the dynamics of agricultural land cover over time and space (Bégué et al., 2018). A large amount of freely available satellite data from the Landsat and Copernicus programmes allows to disentangle phenological characteristics of individual crops. In addressing these complexities, machine learning models have emerged as pivotal tools in remote sensing applications for agricultural monitoring. These models are particularly effective for tasks such as the classification of crop types and land cover (Blickensdörfer et al., 2022). Ensemble learning techniques, such as Random Forests (RF), have gained prominence due to their ability to handle high-dimensional datasets, including spectral-temporal metrics derived from optical and radar imagery. RF models are robust against overfitting and can effectively manage heterogeneous input feature splits (Belgiu & Drăguț, 2016), making them well suited for complex datasets like those used in this study. By leveraging these capabilities, machine learning enables the integration of multi-sensor data to capture the subtle phenological differences and compositional variability of ICs, ultimately enhancing classification accuracy and adaptability to real-world conditions. While this has enabled the production of high-quality crop type maps (Blickensdörfer et al., 2022; d’Andrimont et al., 2021; Griffiths et al., 2019; Johnson, 2019), the current methods focus on the detection of one crop during the growing season, which is usually the main crop. The need for separation of crop types in greater detail is often emphasised but has been addressed by comparatively few remote sensing studies (Bégué et al., 2018). Mapping approaches of ICs are particularly aggravated by data gaps in optical data due to cloudy winter months during the growing season and a lack of reliable reference data (Fendrich et al., 2023; Najem et al., 2024; Schulz et al.,

2021). Missing information and an uneven distribution of data during the growing season highlight the need to integrate data from different satellite sensors and test compositing techniques (Gao et al., 2020; Najem et al., 2024). Furthermore, the similarity of ICs to some main crops, their highly variable seeding times and differing species compositions challenge a distinction of ICs from other cultivated crop types.

Here, we aim to address these gaps by mapping ICs using Spectral-Temporal Metrics (STMs) derived from combined optical and Synthetic-Aperture Radar (SAR) satellite imagery, and incorporating ground-truth data from the field. STMs have been widely used in land-cover mapping and are often preferred to e.g. best-pixel composites or single observations from optical sensors as they provide reliable information for periods of low data coverage (Frantz et al., 2023; Müller et al., 2015; Pflugmacher et al., 2019). By summarising the spectral distribution and temporal variability of a pixel over a specified period, STMs capture important phenological information. Furthermore, integration of SAR data, which unlike optical images are not affected by atmospheric conditions can be beneficial, considering data gaps during the main growth period of ICs (Jennwein et al., 2022; Meroni et al., 2021). This motivates the following research questions:

- How do STMs of combined optical satellite imagery of high spatial resolution (10–30 m) and SAR-based metrics account for the spectral-temporal characteristics of ICs?
- How does the integration of different input features, including optical metrics, SAR metrics, and main crop information, impact the classification accuracy of intermediate crops?
- To what extent do the obtained results align with official agricultural statistics and may contribute to long-term monitoring of agricultural systems?

The paper is structured as follows: Section 2 introduces the study area, Section 3 describes the data and preprocessing steps, Section 4 outlines the methods used for classification, Section 5 presents the results, and Section 6 discusses the findings and concludes the study.

Study area

The study area is located in north-eastern Germany and comprises the federal states of Berlin and Brandenburg (Figure 1). Its area of approximately 30000 km² is predominantly covered with agricultural land (~44%), of which 84% is arable land (MLUK, 2021b). Brandenburg is characterised by a warm temperate climate with annual precipitation totals averaging 558 mm and a mean annual air temperature of 9.2 °C (DWD, 2019). With increasing continental influence, the southern and south-eastern regions of

Brandenburg experience a slightly warmer and drier climate. Soils within the study area evolved from quaternary loose sediments (MLUK, 2020) and are comprised of sandy to sandy-loamy substrates that vary in their suitability for agricultural use (Hanff & Lau, 2021; Wolff et al., 2021). Sandy soils in the outwash plains and meltwater valleys provide low water retention capacity and soil fertility. They are mostly used as grassland or non-agricultural land. Instead, sandy-loamy soils found on plateaus of young and old moraines (e.g. Uckermark, Fläming Heath) are more fertile and better suited for crop cultivation. Other intensively used cropland sites are situated on drained floodplain soils along rivers (e.g. Oderbruch).

Phenological stages of commonly grown main crops in the study area were obtained from regular field observations reported by the German Meteorological Service (Deutscher Wetterdienst, DWD; Figure 2). Different spring and winter crops are grown, with silage maize, winter rye and winter wheat occupying most of the area (Amt für Statistik Berlin-Brandenburg, 2021). Certain crop types, e.g. sugar beet, are grown less frequently as they require very good soil qualities. For some crops, phenological data were unavailable, e.g. for potatoes, which show highly variable sowing (March – May) and harvesting (June – October) dates.

To ensure good growing opportunities, ICs should be sown right after harvesting the main crop and no later than by the end of September (Schmidt & Gläser, 2014). Accordingly, two main cultivation windows for ICs can be identified. If grown after an early-harvested main crop (e.g. winter barley, early potatoes), ICs may be sown from the beginning of July. Fast-growing ICs may then be ploughed before a succeeding winter crop, whereas slow-growing ICs remain on the field until next year's spring crop is sown. The latter are referred to as winter ICs and are most common. They can either serve as forage if winter-hardy or leave frostbitten crop residues in spring. We will consider June 2021 to March 2022 to map different ICs.

Data

Optical satellite data

The growing seasons of ICs require the use of potentially cloudy and noisy winter imagery. We therefore combined different optical satellite data acquired between 1 June 2021 and 31 March 2022. Our analysis was based on Landsat 7 (Enhanced Thematic Mapper Plus, ETM+), Landsat 8 (Operational Land Imager/Thermal Infrared Sensor, OLI/TIRS), Landsat 9 (Operational Land Imager-2/Thermal Infrared Sensor-2, OLI-2/TIRS-2) and Sentinel-2A/B (Multispectral Instrument, MSI) scenes with a maximum cloud cover of 70%. The different sensor swaths overlap laterally, resulting in shorter revisit intervals and higher observation numbers

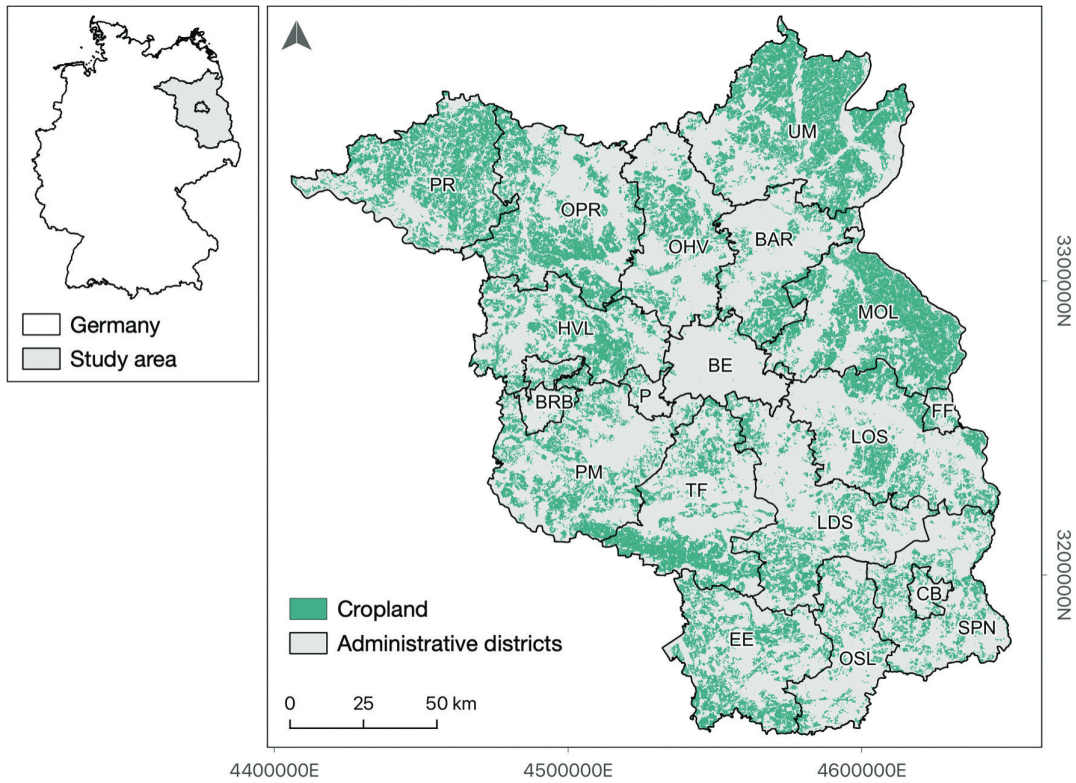


Figure 1. Cropland in Brandenburg and Berlin, Germany (cadastral data ©MLUK, dl-de/by-2-0). BAR: Barnim; BE: Berlin; BRB: Brandenburg an der Havel; CB: Cottbus; EE: Elbe-Elster; FF: Frankfurt (Oder); HVL: Havelland; LDS: Dahme-Spreewald; LOS: Oder-Spree; MOL: Märkisch-Oderland; OHV: Oberhavel; OPR: Ostprignitz-Ruppin; OSL: Oberspreewald-Lausitz; P: Potsdam; PM: Potsdam-Mittelmark; PR: Prignitz; SPN: Spree-Neiße; TF: Teltow-Fläming; UM: Uckermark (©GeoBasis-DE/BKG (2022), dl-de/by-2-0).

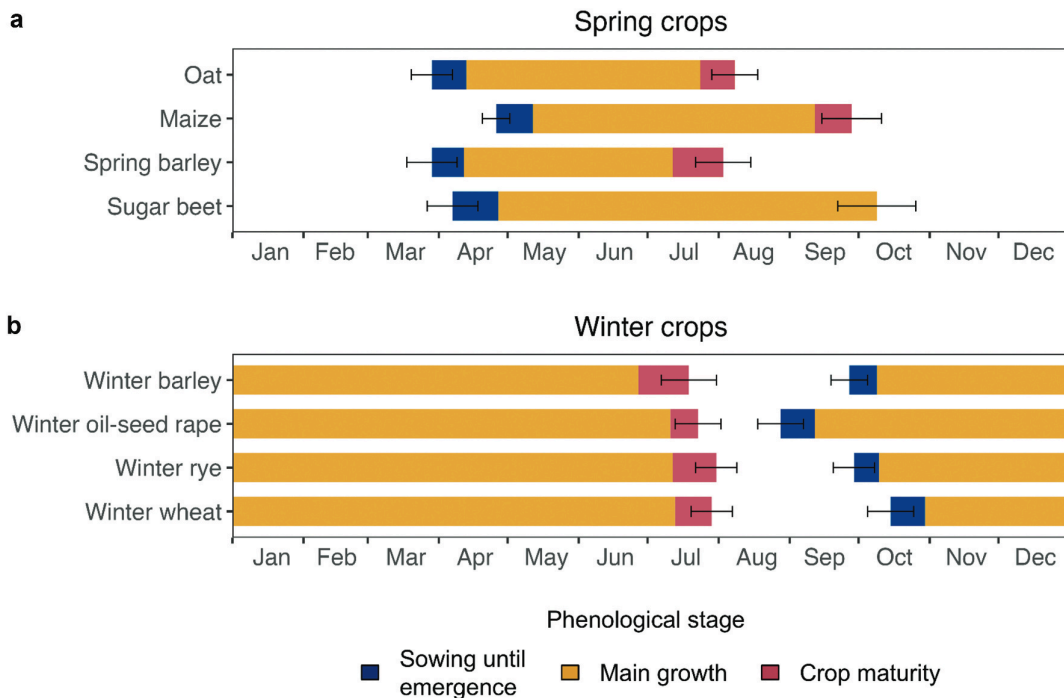


Figure 2. Average vegetation periods of common main crops in Brandenburg based on phenological data from 2019–2021 (DWD Climate Data Center CDC, 2021, 2022). The error bars display standard deviations for the observed sowing and harvesting dates. (a): Phenology stages of commonly grown spring crops. (b): Phenology stages of commonly grown winter crops.

(Li & Roy, 2017). This allows to generate denser time series which are beneficial for remote sensing applications that require reconstructing the phenological behaviour of plants, specifically agricultural land cover mapping. As of 1 January 2022, Landsat 7 imagery was excluded from the analysis due to its official replacement by Landsat 9 which offers improved radiometric and geometric quality. Satellite imagery was obtained and processed using the Framework for Operational Radiometric Correction for Environmental monitoring (FORCE). FORCE provides cloud masking, radiometric correction and higher-level processing modules in a data cube structure (Frantz, 2019). All accessed satellite data were quality masked to exclude clouds, cloud shadows, snow, no data and saturated or sub-zero reflectance. The native Landsat (30 m) and Sentinel-2 (20 m) bands were resampled to obtain a spatial resolution of 10 m, while the 10 m native Sentinel-2 bands remained unchanged. Instead of using the raw bands, we selected five vegetation indices for our analysis – namely Soil-adjusted vegetation index (SAVI), Normalised difference tillage index (NDTI) and Tasseled Cap Brightness, Wetness and Greenness (Table 1).

Sentinel-1 Ground Range Detected (Interferometric Wide swath mode (IW), Processing level 1 high-resolution products) data were integrated in the study. To ensure optimal quality and consistency, the SAR data underwent several pre-processing and filtering processes. Initially, Sentinel-1 observations were filtered based on their acquisition mode – either descending or ascending – and their relative orbit. Additionally, data with extreme incidence angles, which often result in significant noise, were masked. We further pre-processed the data using a Gamma Map Filter (Lopes et al., 1990) with 5×5 kernel size to reduce speckle noise. The cross-polarisation ratio (VH/VV) was computed from the pre-processed data to augment our optical data time series.

Field data

A field campaign focusing on the detection of ICs was conducted throughout different agricultural landscapes in Brandenburg. Field data were collected from late August 2021 until April 2022, with the majority of records reported at the end of September and in November 2021. A total of 759 fields were reported, of which 509 were covered with ICs. The documented variables include the observed crop species according to Table 2, corresponding coordinates, date and additional comments, e.g. on vegetation density and suitability for subsequent analysis. The crop species were explicitly noted to keep track of which IC types and non-IC crop types were sufficiently covered by the field campaign. The samples were then assigned to their corresponding target class for classification (cf. Table 2). The target class “other crop types” refers to non-ICs observed in the study area during the campaign. This includes fallow land, fodder or

Table 1. List of indices used in the study.

Index	Name	Formula	Reference
SAVI	Soil-adjusted vegetation index	$SAVI = ((NIR - RED) / (NIR + RED + L)) * (1 + L)$	Huete (1988)
NDTI	Normalised difference tillage index	$NDTI = (SWIR1 - SWIR2) / (SWIR1 + SWIR2)$	van Deventer et al. (1997)
TCB	Tasseled cap brightness	$TCB = 0.2043 * BLUE + 0.4158 * GREEN + 0.5524 * RED + 0.5741 * NIR + 0.3124 * SWIR1 + 0.2303 * SWIR2$	Crist (1985)
TCG	Tasseled cap greenness	$TCG = -0.1603 * BLUE - 0.2819 * GREEN - 0.4934 * RED + 0.7940 * NIR - 0.0002 * SWIR1 - 0.1446 * SWIR2$	Crist (1985)
TCW	Tasseled cap wetness	$TCW = 0.0315 * BLUE + 0.2021 * GREEN + 0.3102 * RED + 0.1594 * NIR - 0.6806 * SWIR1 - 0.6109 * SWIR2$	Crist (1985)

Table 2. Observed crop types and their corresponding training and validation sample sizes obtained after initial cleaning of the field data and balancing of the training dataset.

Target class	Crop type	Training sample size	Validation sample size
Other crop type	Fallow land	5	1
	Fodder or sugar beet (<i>Beta vulgaris</i> ssp. <i>vulgaris</i>)	6	1
	Oil-seed rape (<i>Brassica napus</i>)	104	40
	Vegetables	4	0
	Volunteer oil-seed rape (<i>Brassica napus</i>)	6	1
	Volunteer cereals	59	12
	Weeds	18	5
Intermediate crop	Winter cereals	70	20
	Lucerne (<i>Medicago sativa</i>)	15	8
	Black oat (<i>Avena strigosa</i>)	7	2
	Buckwheat (<i>Fagopyrum esculentum</i>)	3	1
	Clover (<i>Trifolium incarnatum</i> , <i>Trifolium pratense</i> , <i>Trifolium resupinatum</i> , <i>Trifolium repens</i>)	4	1
	Lupine (<i>Lupinus</i> spp.)	5	1
	Oil radish (<i>Raphanus sativus</i> var. <i>oleiformis</i>)	6	1
	Pea (<i>Pisum sativum</i>)	5	1
	Phacelia (<i>Phacelia tanacetifolia</i>)	18	8
	Ramtil (<i>Guizotia abyssinica</i>)	2	0
	Rye (<i>Secale cereale</i>)	5	1
	Ryegrass (<i>Lolium perenne</i>)	2	0
	Seed mixture with legumes	133	75
	Seed mixture without legumes	76	41
	Serradella (<i>Ornithopus sativus</i>)	1	0
	Sunflower (<i>Helianthus annuus</i>)	4	1
	Turnip rape (<i>Brassica rapa</i>)	8	2
	White mustard (<i>Sinapis alba</i>)	36	15
	Other sole crops (<i>Cannabis spec.</i> , <i>Helianthus tuberosus</i> , <i>Sisymbrium spec.</i> , <i>Sorghum spec.</i>)	5	0

sugar beet, oil-seed rape, vegetables, volunteer crops, weeds, and winter cereals. These categories encompass a mix of unmanaged vegetation, residual crops from previous harvest, or main crops that were still or already present in the fields. The target class “intermediate crops” refers to crops that were identified as ICs based on their time of observation and development stage. It is comprised of various sole crops, seed mixtures and potential perennial species. At most sites, pictures were taken to allow retrospective data interpretation. During the mapping process, particular attention was drawn to (i) maintaining a certain distance between fields of the same crop type to avoid spatial autocorrelation, (ii) capturing rarely observed crop species, and (iii) ensuring a good representation of within-class variability.

Training and validation sample

After a data cleaning procedure that minimised sampling near headlands, which tend to be unrepresentative due to compaction and external inputs (Sunoj et al., 2021). (Appendix A), and avoided mixed pixels along field boundaries, a final selection of 734 fields was retained for analysis. As the information on ICs captured by IACS data was considered too incomplete for validation purposes, we split the field points into training and validation samples. Classes with limited representation in the field data were included in the training dataset as they reflect the real-world complexity and diversity of ICs present in the study area. Not considering these crop types for training would have oversimplified the classification task and reduced the ecological validity of the models. However, the small

amount of field samples of certain crop types (e.g. ryegrass) did not allow to include them for both training and validation. Consequently, IC types that were represented poorly in the field data were not used for validation since their underrepresentation limited their reliability for assessing model performance on false positives or false negatives. Otherwise, validation samples were manually selected in proportion to their total count in Table 2 without choosing points that had been relocated during the data cleaning, e.g. due to buffer strips or heterogeneous parcels. All remaining points formed the training set which was initially affected by class imbalance. We therefore adjusted the training data by duplicating samples from the class “other crop type” whenever a minimum distance of 300 m to the neighbouring training point of the same field could be maintained (cf. Figure B1). This duplication ensured better balance in the training dataset by addressing class imbalance while maintaining spatial distinctness to avoid introducing spatial autocorrelation. Figure 3 illustrates the spatial distribution of the selected samples indicating the respective target class and their use for training or validation.

Methods

Spectral-temporal metrics and data availability

We defined four different aggregation periods for the computation of seasonal STMs, covering the complete growing season of ICs. The periods were selected with regard to the data availability of optical satellite data (Figure 4) and the expected phenology of crop types in the study area (Table 3).

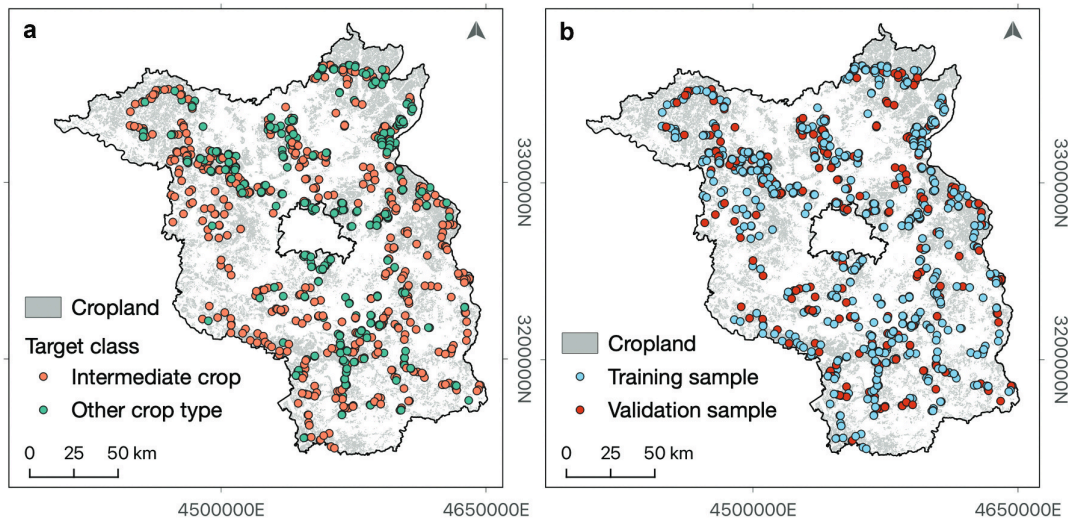


Figure 3. Spatial distribution of ground-truth data gathered in the field (cadastral data ©MLUK, dl-de/by-2-0). (a): Target classes. (b): Training and validation samples.

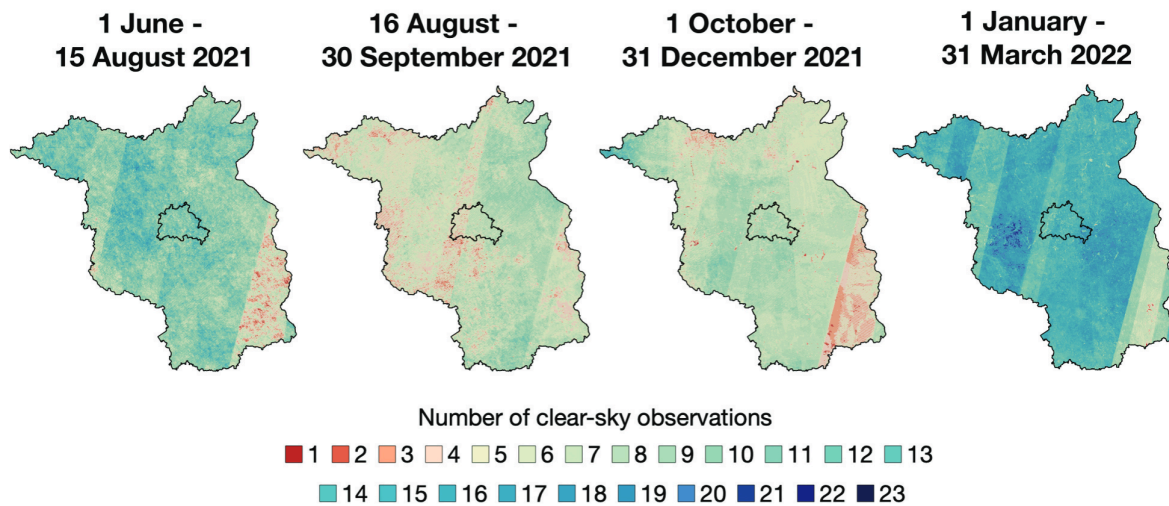


Figure 4. Clear-sky observation counts for each aggregated period (1 June – 15 August 2021: T1; 16 August – 30 September 2021: T2; 1 October – 31 December 2021: T3; 1 January – 31 March 2022: T4).

Table 3. Anticipated crop development within the selected aggregation periods.

No.	Aggregation period	DOY	Expected phenology
T1	1 June 2021–15 August 2021	152–227	Harvest of main crops Sowing and emergence of early ICs
T2	16 August 2021–30 September 2021	228–273	High biomass of early grown ICs Growth of weeds and volunteers Sowing and emergence of winter ICs Sowing of winter main crops
T3	1 October 2021–31 December 2021	274–365	Harvest of remaining main crops (maize, sugar beet) Fast development of ICs Slow development of other crop types
T4	1 January 2022–31 March 2022	001–090	Winter dormancy of main crops and winter-hardy ICs with continuing growth in spring Frost damages of most ICs

The number of valid observations per pixel, referred to as clear-sky observations (CSOs), varied strongly across the defined periods and across the study area, depending on orbit overlaps and

cloudiness. During summer, it was possible to select shorter aggregation periods with average numbers of observations yielding 9.7 for T1 and 6.2 for T2. T3 and T4 include winter months and therefore cover longer

time spans to ensure continuous data coverage. Nevertheless, a large part of the study area depicted less than five CSOs per pixel between 1 October 2021 and 31 December 2021, due to missing orbit overlap, the sensor failure of Landsat 7 and a higher incidence of clouds during autumn and winter. The obtained data densities for this period (T3, average CSOs = 6.5) provide a more typical representation of the winter condition, compared to the second winter period (T4, 1 January 2022–31 March 2022) which depicts an unusually high number of 14.3 CSOs on average due to exceptional cloud scarcity. Given the limited number of cloud-free observations, particularly during the winter months, compositing was employed to ensure sufficient spatial and temporal coverage under suboptimal conditions. By focusing on STMs that summarise phenological trends over these defined periods, we mitigated the potential impact of using observations from different dates and reduced the influence of short-term variability.

STMs were computed from each index or band (NDTI, SAVI, TCB, TCG, TCW, VV, VH, VH/VV). For optical data, minimum (MIN), maximum (MAX), range (RNG), median (Q50), standard deviation (STD) and interquartile range (IQR) values were derived for each period – adding up to 120 features. A reduced number of features was produced for the Sentinel-1 bands including MAX, MEAN, Q50 and STD metrics.

Classification models

In this study, the RF ensemble learning method (Breiman, 2001) was chosen for classification due to its robustness against overfitting and capability to handle high-dimensional data typical of remote sensing. It has been widely used in classification tasks using remotely sensed data (Blickensdörfer et al., 2022; Pelletier et al., 2016; Sherrie; Wang et al., 2019).

Several scenarios were tested based on time steps, optical and SAR data input (Table 4) to evaluate the relative importance of different input features. The models furthermore considered information on the scheduled main crop in 2021 as declared in the IACS (MLUK, 2021a) as input (cf. Figure B1). Each model

was based on 500 decision trees, and the square root of the total number of features was randomly sampled at each split (Belgiu & Drăguț, 2016). The models were applied to produce a binary crop type map across actively managed cropland sites in the study area (MLUK, 2021b). Post-classification, the minimum mapping unit (MMU) of the obtained maps was changed from a single pixel to the IACS polygon level. To achieve this, a frequency analysis of all pixel classes within a given parcel was performed. The most frequent class was then assigned to the corresponding field, and the aggregated maps were masked to match the cropland areas that were used for prediction. Following the classification, feature importance was extracted based on the mean decrease in Gini impurity. This metric evaluates the contribution of each feature to reducing classification uncertainty by assessing its role in splitting decision nodes across the trees in the RF model. Feature importance was calculated for all input variables, including STMs derived from optical and SAR data, to determine the most significant drivers for distinguishing ICs from other crop types.

Accuracy assessment

We performed a statistical accuracy assessment by comparing the validation data to the different map classifications at pixel and field level. The resulting confusion matrices were used to calculate common coefficients, including Overall Accuracy (OA), Producer's Accuracy (PA) and User's Accuracy (UA).

We used the map with highest OA to estimate the area of each target class and compared it to agronomic statistics. Official information on agricultural areas including ICs was obtained from agricultural structure surveys conducted in 2010, 2016 and 2020 (Destatis, 2011, 2017, 2021). In contrast to our study, perennial crops – which by definition are no ICs – were specifically excluded from the surveys. We therefore subsequently excluded lucerne and clover fields that were reported as main crops in the 2021 IACS data from the area assessment. Thereby we improved the comparability to the survey results while not changing the map design.

Table 4. Classification models. Each model was based on a different set of input features including spectral-temporal metrics from optical data, spectral-temporal metrics from SAR data or both. All models additionally included information about the cultivated main crop in 2021.

Model	Input features	Number of features
Optical	5 vegetation indices $\times$ 4 time periods $\times$ 6 metrics + main crop	121
SAR	3 bands $\times$ 3 time periods (T2–T4) $\times$ 4 metrics + main crop	37
Combined	5 vegetation indices $\times$ 4 time periods $\times$ 6 metrics + 3 bands $\times$ 3 time periods (T2–T4) $\times$ 4 metrics + main crop	157

Table 5. Validation results for the obtained maps. For classifications at pixel level, we compared the validation sample to the class assigned to the pixel containing the validation sample. For field level, we compared the validation sample to the most frequent class within each corresponding field.

Classification	Overall accuracy in %	Producer's accuracy in %		User's accuracy in %	
		Intermediate crop	Other crop type	Intermediate crop	Other crop type
Optical, pixel level	85.36	94.97	66.25	84.83	86.89
Optical, field level	92.47	96.23	85.00	92.73	91.90
SAR, pixel level	78.48	93.04	49.37	78.61	78.00
SAR, field level	83.68	93.08	65.00	84.09	82.54
Combined, pixel level	90.30	96.20	78.48	89.94	91.18
Combined, field level	92.89	96.23	86.25	93.30	92.00

Results

Model comparison

The fitted models achieved different accuracies for distinguishing ICs grown between summer 2021 and spring 2022 from other crop types in the region (see Table 5). Overall accuracies ranged from 78.5% to 92.9%, depending on the different input features used and the chosen MMU.

While optical data alone yielded high classification accuracies, the independent use of SAR data led to less accurate results. Highest accuracies were attained for the combined use of optical and SAR data, indicating that SAR data can effectively improve the classification. Moreover, aggregation to field level consistently improved OA and class-wise accuracies across all models. This particularly reduced the omission error of other crop types by the *Optical* and SAR models. The improvement in accuracy, however, was less pronounced for the *Combined* model, suggesting that the model provides more robust estimates of the target classes. Despite the overall enhancement, aggregation to field level occasionally introduced artefacts, e.g. if the declared IACS field boundaries differed from the

actual cropping pattern. The applied majority vote would then result in an under- or overestimation of the IC area of the respective field.

In general, features from T3 and T4 were found to have the largest impact on all classifiers (Figure 5). In the *Optical* model, features from T2 proved to be valuable inputs when providing information on low or high biomass levels (e.g. TCG-MIN/MAX, SAVI-MIN/MAX), while T1 features were least informative. Most optical indices – except for TCB – demonstrated high importance with a particular relevance of their MIN, MAX and Q50 values. These trends were also reflected in the *Combined* model where feature importances of optical input variables were usually higher than importances of SAR metrics. Nevertheless, the highest-scoring SAR metrics (VV/VH from T3) had a significant impact on the *Combined* model's performance. In contrast, SAR data from T1 introduced noise leading to a decrease in accuracy by over 3% and were excluded from the final models. Compared to the satellite data, the inclusion of the cultivated main crop in 2021 as input feature provided limited value and resulted in moderate feature importances across all models.

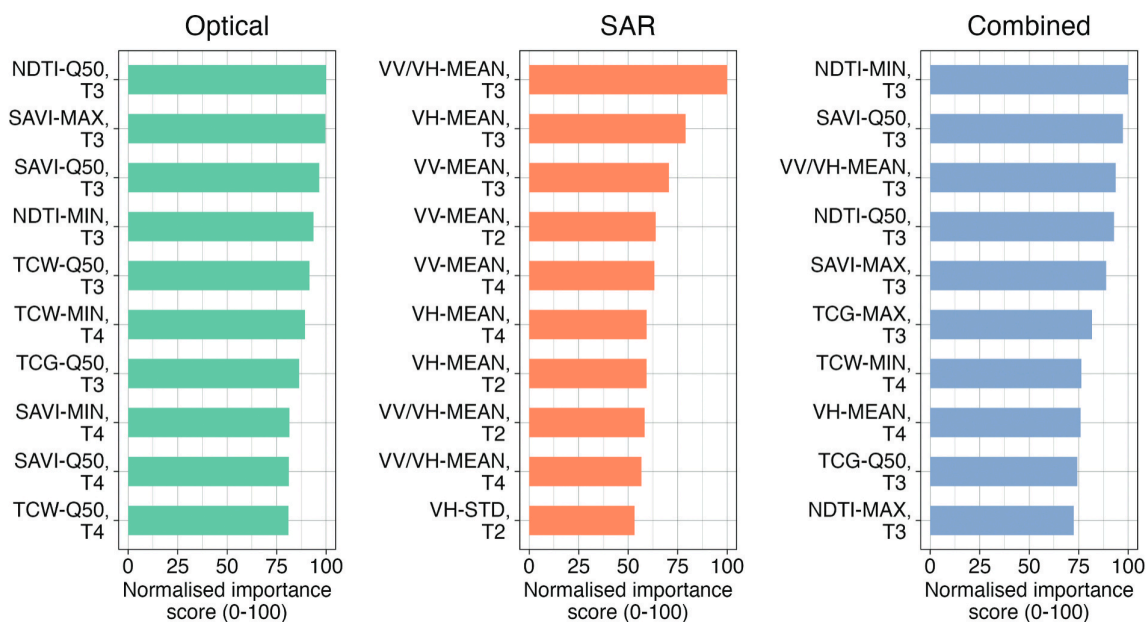


Figure 5. Normalised feature importance of the fitted models. Shown are the ten most important variables of each tested input set. The features are labelled by their index or band, statistical metric and aggregation period.

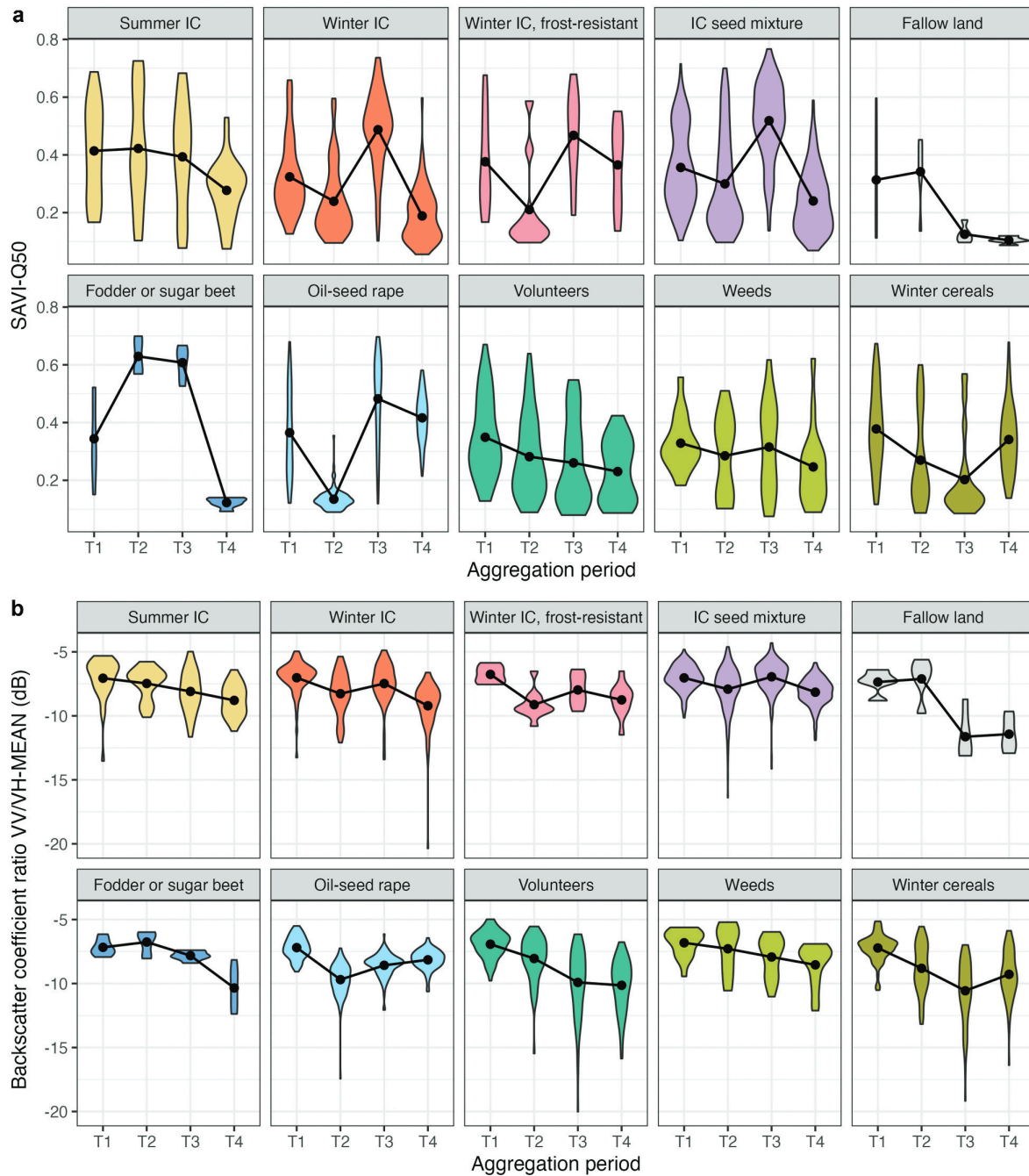


Figure 6. Temporal profiles for observed crop groups, considering training samples only. The black lines represent average phenological trends over the aggregation periods used to calculate the spectral-temporal metrics, the violins display underlying value distributions. (a): Optical metric (median of the soil-adjusted vegetation index: SAVI-Q50). (b): SAR metric (average cross-polarisation ratio: VV/VH-MEAN).

Representation of phenological patterns

The STMs used in this study were computed for periods that specifically aim to capture the anticipated phenology of ICs in the study area. In an exemplary way, SAVI-Q50 and VV/VH-MEAN profiles were examined to assess the actual representation of phenological patterns (Figure 6). For this purpose, some of the crop types reported in the field were grouped to facilitate visualisation. The grouping was chosen as

a compromise between interpretability and detail, allowing to identify potential sources of misclassification that would otherwise be masked by the great heterogeneity within the two target classes.

Optical and SAR metrics often displayed similar patterns and were able to detect phenological differences between groups (Figure 6). The largest differences were observed for T2, T3 and T4, corresponding to the feature importance results (Figure 5). ICs were